

Liquid-Liquid Extraction

Pre-Lab Plan Unit Operations Lab #5

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Group 2, Tuesday, PM section

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1. Experimental Objective (5 pts)

The objective of the experiment is to run a three-stage mixer-settler liquid-liquid extraction system and take acetic acid out of a white oil feed using water as the extracting solvent. Additionally, the experiment is used to evaluate the impact of the agitator speed on extraction efficiency and to determine the impact of mixing intensity on the number of effective equilibrium stages in the system.

2. Experimental

A feed mixture was made that had about 1.0 wt. acetic acid in white oil and mixed thoroughly in the feed tank. Standardized NaOH solution and phenolphthalein were used to calculate the first concentration of acetic acid through a titration reaction with phenolphthalein. Water was then added to the three-stage mixer-settler unit to stabilize the liquid levels and then the agitators were adjusted to a desired rate, and the oil feed was turned on. The flowrates of water and oil were regulated to ensure solvent to feed ratio of 2:1. Once the steady state had been achieved, both aqueous and oil phases of every stage were sampled and analyzed by titration to ascertain the concentration of acetic acid. This process was carried out at three agitator speeds, and flowrates remained at a constant level.

2.1 Apparatus (5 pts)

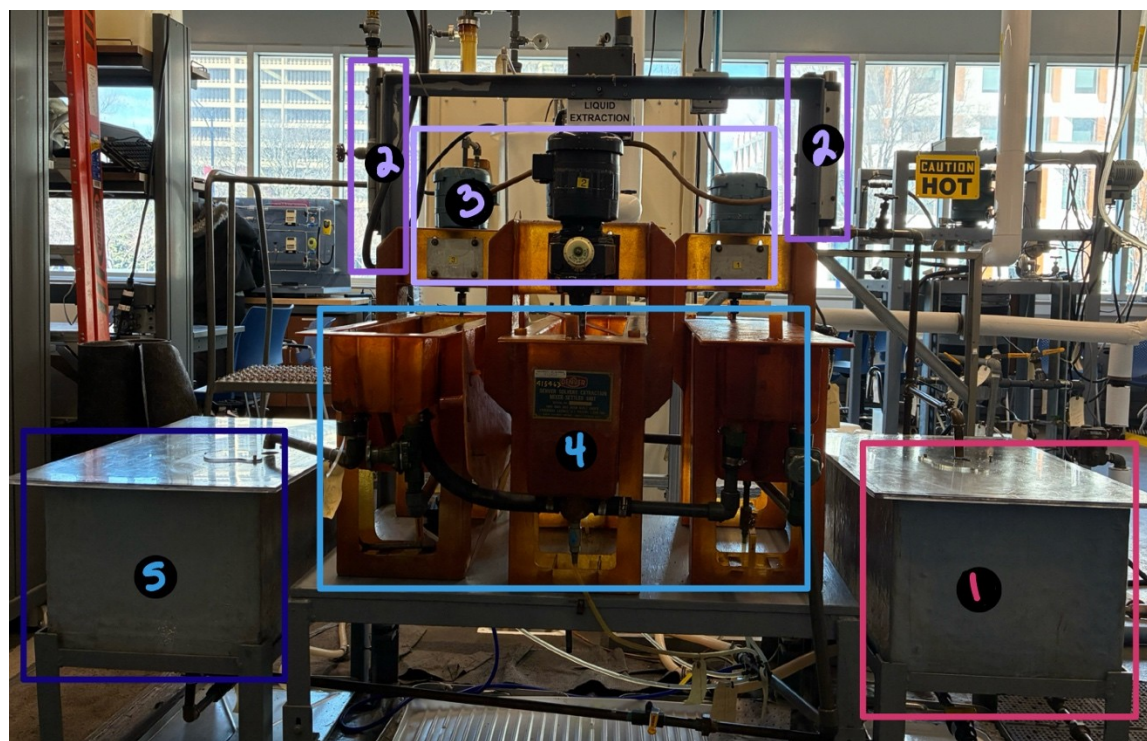


Figure 1: Apparatus with color coded and numbered labeling.

Table 1: Apparatus parts correlating to Figure 1.

Number/ Color	Apparatus Part
1 – Pink	Mineral oil with acetic acid reservoir – system “inlet”.
2 – Purple	Manometers for both water (LHS) and oil (RHS).
3 – Light purple	Mixers (3) for individual stages.
4 – Light Blue	Stripping stages (3).
5 – Dark Blue	Finished product reservoir – system “outlet”.
Not Pictured	Pump

2.2 Materials and Supplies (5 pts)

Table 2: Materials and supplies used in experiment and their purpose.

Materials/Supplies	Purpose
Mineral Oil	Component that acetic acid is separated from.
Acetic Acid	Added to mineral oil to be separated out.
Water	Used for stripping.
Tachometer	Measuring the speed of the mixers.
Burette	Used for titration.
Beaker/Erlenmeyer Flask	Used to hold sample/titrate sample in.
Phenolphthalein	Titration indicator.
NaOH	Used in titration to determine acid within sample concentration.
Parafilm	Used to cover samples until they can be titrated.

2.3 Experimental Plan (10 pts)

- Inspect the three-stage Denver mixer-settler apparatus before operation to identify the function of each stage, including the mixer, settler, agitator, rotameters, and sampling points. Verify that all reflux valves remain fully closed for the duration of the experiment.
- Confirm that sufficient white oil is present in the feed tank (~25 gallons).
- Determine whether the recycled white oil already contains any residual acetic acid by taking an initial sample and titrating it with standardized NaOH. Use this information when preparing the final feed mixture.
- Prepare the feed as ~1.0 wt% acetic acid in white oil. Mix thoroughly in the feed tank by circulating the contents with the centrifugal pump in a closed loop, then withdraw a representative sample to determine the exact initial acetic acid concentration. Keep the feed tank covered to minimize acetic acid evaporation.
- Prepare for composition analysis by using standardized NaOH solution and phenolphthalein indicator for titration of both oil-phase and water-phase samples. Select the NaOH normality that gives a clear and measurable endpoint for the expected acid concentration range.

- Establish the water circuit first by introducing water into the extraction unit and adjusting the solvent flowrate to about 0.67 GPM using the rotameter.
- Turn on all three agitators and set them to the first selected mixing speed using a tachometer. Choose speeds within the acceptable operating range of 350 to 1000 rpm so that phase circulation is stable but excessive entrainment is avoided.
- Once the water circuit is stable, begin the oil feed by directing the acetic acid/white oil mixture from the feed tank to the extraction unit. Adjust the oil flowrate so that the solvent-to-feed ration is at least 2:1. A typical starting condition is about 0.34 GPM oil with 0.67 GPM water.
- Fine-tune the oil and water flowrates by trial and error until stable two-phase levels are maintained in all three stages. After stable operation is achieved, keep both flowrates constant throughout the remainder of the experiment.
- Measure and record the oil flowrate, then estimate how long the feed supply will last so that all three runs can be completed within the available 25 gallons of prepared feed.
- Allow the system to operate until oil first reaches the product tank. Record this time as an estimate of the oil residence time through the extraction system.
- At steady operation for the first agitator speed, quickly collect samples of both the raffinate phase (oil) and the extract phase (water) from each of the three stages. Cover samples immediately to minimize evaporation losses prior to analysis.
- Titrate each sample to determine the acetic acid concentration in both phases. For oil samples, use a stirrer or stir bar during titration to improve contact between nonaqueous sample and the aqueous NaOH titrant.
- Repeat the same sampling and titration procedure for two additional agitator speeds, while holding solvent flowrate, feed flowrate, feed composition, and valve positions constant. This gives a total of three operating runs, each corresponding to a different mixing speed.
- After the final run, leave the oil in the product tank and stages rather than draining it, since the white oil is reused throughout the semester.

The primary independent variable in this experiment is the agitator speed, which will be tested at three different values selected within the allowable range of 350-1000 rpm. A second controlled operating parameter is the solvent-to-feed ratio, which should remain at or above 2.0, with typical operating conditions of about 0.67 GPM water and 0.34 GPM oil. The dependent variables are the acetic acid concentration in the extract phase and raffinate phase leaving each stage, usually expressed as Y_A and X_A , respectively. These concentrations will be measured by acid-base titration using standardized NaOH and phenolphthalein indicator. Since samples will be taken from each of the three stages for both phases, each run produces six concentration measurements, allowing concentration profiles to be constructed as a function of stage number.

The experimental design calls for three total runs, one at each agitator speed. The baseline plan includes one steady-state run at each of the three agitator speeds, replicate titrations should be performed for each collected sample when possible to improve analytical reliability. For uncertainty reduction, the feed should be mixed thoroughly before sampling, the water circuit must be established before oil is introduced, all tanks and stages should remain covered to prevent acetic acid evaporation, and samples should be covered with parafilm before titration. In

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addition, the titration endpoint must be read carefully, and oil samples should be stirred during titration because the system involves an aqueous titrant reacting with a nonaqueous sample. The most likely sources of uncertainty will be premature sampling before true steady state is reached, poor endpoint identification, evaporation losses of acetic acid, improper flow selection, and operation at an unsuitable mixing speed.

For statistical treatment, titration results for repeated analyses of the same sample will be averaged, and uncertainty will be reported using a 95% confidence interval when replicate titrations are available. If only single titrations are performed per sample, then formal confidence intervals cannot be calculated from the experimental data alone and uncertainty will be discussed qualitatively based on instrument precision, flow stability, and endpoint interpretation. Goodness of fit for any equilibrium or operating-line correlations will be assessed using percent error, absolute deviation, or R^2 where a linearized relationship is fitted to the data.

The collected data will be correlated with extraction theory in several ways. First, the measured acetic acid concentrations in the extract and raffinate phases will be plotted as functions of stage number for each agitator speed to show how solute transfer progresses through the three-stage cascade. Second, the measured Y_A and X_A values will be plotted against one another and compared with the equilibrium information provided in Appendix I of the lab manual. Third, a McCabe-Thiele analysis will be performed using the equilibrium data to estimate the number of theoretical stages required for the observed separation. Finally, the measured stage compositions will be used to calculate Murphree stage efficiencies and an overall extraction efficiency, allowing direct comparison between actual and ideal stage performance.

Project Safety Evaluation (10 pts)

Process safety considerations for the liquid–liquid extraction experiment primarily involves proper chemical handling, equipment operation, and waste management. Although no significant chemical reactions occur during the process, dilute sodium hydroxide (approximately 0.05 M) is used for titration, which is mildly corrosive and requires appropriate personal protective equipment (gloves, goggles, and lab coat) to prevent skin and eye irritation. Phenolphthalein indicator should also be handled carefully to avoid contact and staining. The mineral oil feed contains approximately 1 wt% acetic acid, which presents minimal hazard beyond a mild vinegar-like odor, however, adequate ventilation should be maintained to prevent discomfort. Proper segregation and disposal of aqueous waste streams into designated containers is critical to ensure compliance with laboratory safety and environmental guidelines.

Mechanical hazards are also important in this experiment due to the use of high-speed mixers in each extraction stage. While the impeller blades present minimal direct contact risk, it is essential that all stages remain properly sealed with lids during operation to prevent splashing or leakage of oil and aqueous phases. Escaping liquid could create slip hazards on the laboratory floor, increasing the risk of injury. Additionally, care should be taken to avoid loose clothing or hair near rotating equipment. Ensuring that all equipment is securely assembled and operated according to procedure will minimize the risk of mechanical failure or accidental exposure.

	Yes/No	
Potential electrical hazards?	Y	If yes, identify conditions: Pump, electric w/ water
Potential mechanical hazards?	Y	If yes, identify conditions: Impeller
Potential pressure hazards?	N	If yes, identify conditions:
Potential temperature hazards?	N	If yes, identify conditions:
Hazardous/toxic chemicals involved?	Y	If yes, hazards involved: NaOH, acetic acid (5wt%)
Gloves required?	Y	If yes, specify type: Nitrile
MSDS reviewed by all team members?	Y	List MSDS sheets reviewed: CAS-No: 130-73-2 CAS-No: 64-19-7, 7732-18-5
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	RT	
Pressure (psia)	Atm	
Safety Controls	yes/no	If yes, Provide description
	Y	System shut off

I have read relevant background material for the Unit Operations Laboratory entitled: "Liquid - Liquid Extraction" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

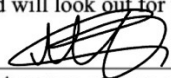
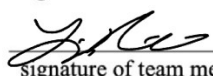
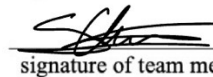
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Figure 2: Safety evaluation chart.